

THE P VERSUS NP PROBLEM RESEARCH BRIEF

Prepared for: theory-of-computation course review

Scope: What the P versus NP problem formally states, why it has resisted proof, and where the research actually stands.

Level: Undergraduate computer science theory. Turing machines, formal languages, big-O notation, polynomial-time reductions, Cook-Levin and NP-completeness are treated as working vocabulary, not rebuilt from zero.

SECTION 01

1. SHORT ANSWER

- P versus NP asks whether every decision problem whose solutions can be verified in polynomial time can also be solved in polynomial time.** P is the class of languages decidable in time bounded by a fixed polynomial in the input length, a definition traced to Jack Edmonds in 1965 [1]; NP is the class whose solutions are verifiable in polynomial time by a deterministic machine [1][4].
Implication: The question is a precise statement about two classes of languages, not a loose claim about which problems feel hard.
- The problem is open in the strong sense that the three general proof techniques the field has are each known to be insufficient, by theorem.** Diagonalization fails because any such proof would relativize, and Baker, Gill and Solovay 1975 showed no relativizing proof can settle the question in either direction [1][3]; combinatorial circuit lower bounds run into natural proofs [1]; arithmetization runs into algebrization [3]. **Implication:** A correct proof must use a technique that is currently unknown, not a sharper version of an existing one.
- The barriers are results about proof methods, not evidence about the answer.** Aaronson and Wigderson show that almost all the major open problems, P versus NP included, will require non-algebrizing techniques, and that algebrization explains exactly where progress stopped, for instance why superlinear circuit lower bounds exist for PromiseMA but not for NP [3]. **Implication:** "We cannot prove it with these tools" is a theorem; "P is probably not NP" is a belief, and the two should never be run together.
- The state of research is candidly described by the field as stalled rather than converging.** Fortnow's 2009 survey opens by saying the article could be written in two words, "Still open," and reports that circuit complexity and other approaches have stalled with little reason to expect a separation soon [1]; his 2021 to 2022 follow-up states the resolution remains far out of reach [2]. **Implication:** No current programme has a credible near-term path, and the best-known long-range one, Geometric Complexity Theory, is self-described as a multi-decade effort.
- The consequences of $P = NP$ are far broader than breaking cryptography, and the consequences of $P \neq NP$ are far narrower than securing it.** Fortnow argues the gains from $P = NP$ would dwarf the loss of public-key cryptography, while separately noting that assuming $P \neq NP$ is not enough to get public-key protocols, which need strong average-case assumptions [1]. **Implication:** $P \neq NP$ is a necessary but not sufficient condition for modern cryptography, which is the single most load-bearing point in this brief for a student.

- **Practice has moved a long way without the theory moving at all.** Fortnow's 50-year retrospective records dramatic advances in algorithms and hardware that let us tackle many NP-complete problems while making little progress breaking cryptographic systems, and names the resulting regime "Optiland" [2]. **Implication:** Worst-case intractability and practical intractability are different claims, and conflating them is the most common error in applying this theory.

SECTION 02

2. DEFINITIONS AND NOTATION

This brief writes the negation as $P \neq NP$ rather than with a struck equals sign, and writes class names in plain capitals. Where the field uses more than one convention, the alternative is named in the third column.

Term	Definition	Notation used here
P	The class of decision problems (languages) solvable by an algorithm within a number of steps bounded by a fixed polynomial in the input length [1]. The class contains problems solvable in polynomial time [4].	P . Rendered as bold or sans-serif capitals in typeset sources.
NP	The collection of problems that have efficiently verifiable solutions [1]. Equivalently, membership in NP means any proposed solution can be verified in polynomial time by a deterministic computer [4].	NP . The name abbreviates "Nondeterministic Polynomial-Time" [1], not "non-polynomial".
The P versus NP question	Whether every problem with an efficiently verifiable solution also has an efficiently computable one [1]. Whether P equals NP remains the central open question in theoretical computer science [4].	$P =? NP$
Polynomial-time reduction	A function, computable in polynomial time, mapping every instance of A to an instance of B with the same yes/no answer; if it exists, a fast algorithm for B yields a fast algorithm for A [4].	$A \leq_p B$
NP-hard	Every other problem in NP reduces to it in polynomial time [4].	$NP\text{-hard}$
NP-complete	In NP and NP-hard simultaneously; the hardest problems inside NP, in the precise sense that a polynomial-time algorithm for any one of them yields polynomial-time algorithms for all problems in NP [4][1].	$NP\text{-complete}$, abbreviated NPC in some texts.
Relativization	A proof relativizes if it still works when every machine involved has	Oracle machines written P^A , NP^A for oracle A [3].

Term	Definition	Notation used here
	access to the same additional information, modelled as an oracle [1].	
Algebrization (algebraic relativization)	Relativizing a class inclusion while giving the simulating machine access not only to an oracle A but also to a low-degree extension of A over a finite field or ring [3].	`algebrize`. Aaronson and Wigderson use both "algebrization" and "algebraic relativization" for the same notion.
P/poly	The nonuniform circuit class appearing in the separations Aaronson and Wigderson catalogue, for example `MAEXP` not-subset-of `P/poly` and the open `NEXP` versus `P/poly` [3].	`P/poly`

SECTION 03

3. KEY FINDINGS BY TOPIC

3.1 The formal statement is about languages and machines, not about difficulty

The question is fully formal: it asks whether two precisely defined classes of languages coincide, and the informal "easy to check implies easy to find" gloss is a translation of that, not the statement itself.

The formalization rests on two definitions that the retrieved sources state consistently.

- P is the class of decision problems solvable within a number of steps bounded by a fixed polynomial in the input length, a notion of "efficient computation" that Jack Edmonds proposed in 1965 alongside his matching algorithm [1].
- NP is the collection of problems whose solutions are efficiently verifiable; membership means a proposed solution can be checked in polynomial time by a deterministic machine [1][4].
- $P = NP$ therefore means that for every problem with an efficiently verifiable solution, we can also find that solution efficiently [1].
- The Clay Mathematics Institute named it one of the seven Millennium Prize Problems in 2000 and offers a million-dollar prize for a proof either way [1][4].

On the official problem description. The source of record is Stephen Cook's official problem description written for the Clay Mathematics Institute. This run identified that document but never opened it (Section 7). Its existence and authorship are independently corroborated here by an unusual artifact: arXiv administratively withdrew submission 1001.3816, titled "The P versus NP Problem", with the notice "This article was plagiarized directly from Stephen Cook's description of the problem for the Clay Mathematics Institute" [6]. That record carries no technical content of its own and is cited here only for that provenance point.

3.2 Cook-Levin is what turns one problem into all of them

NP-completeness is the mechanism that makes the single question P versus NP decide the fate of thousands of separate problems at once, and it is why a proof for any one NP-complete problem settles everything.

The retrieved sources give a consistent account of the history and the mechanism.

- The concept originates with Stephen Cook's 1971 proof and Leonid Levin's independent 1973 result, together known as the Cook-Levin theorem; they showed the Boolean satisfiability problem (SAT) is NP-complete [4].
- Cook presented the paper, "The Complexity of Theorem-Proving Procedures", in Shaker Heights, Ohio in early May 1971 [1]; Fortnow's retrospective dates the introduction to May 4, 1971 [2].
- Cook, Levin and Richard Karp developed the initial theory of NP-completeness, work that generated multiple ACM Turing Awards [1].
- Because reductions compose, once SAT was known to be NP-hard later researchers only needed to reduce from an already-proven NP-complete problem rather than from all of NP directly; Karp's 1972 paper demonstrated 21 NP-complete problems, and the list now runs to thousands [4].

Shape of the reduction

IEEE's summary reports that in Boaz Barak's *Introduction to Theoretical Computer Science*, the Cook-Levin proof establishes 3-SAT as NP-hard by showing every NP problem reduces to it, with the reduction built as a chain through NANDSAT and 3NAND [4]. That is one textbook's packaging of the proof, not the only one; the classical presentation encodes an accepting computation of a nondeterministic machine as a Boolean formula. A separate retrieved preprint records that Cook showed SAT restricted to exactly three variables per clause remains NP-complete [11], available here only as a preview snippet.

REQUIRED ACTION: Check which presentation of the Cook-Levin reduction your own course uses, the tableau or formula encoding of a nondeterministic computation or the NANDSAT chain, because the two look very different on an exam even though they prove the same theorem.

3.3 Why it is open: three techniques, three theorems saying they cannot work

The reason P versus NP has resisted proof is not that nobody has tried hard enough; it is that each of the field's general lower-bound techniques has been proved incapable of settling it.

Diagonalization and the relativization barrier

- Diagonalization goes back to Cantor's 1874 proof that the reals are uncountable, and Turing used a similar technique to show the Halting problem is not computable [1].
- It fails here for two reasons: it requires simulation, and we do not know how a fixed NP machine can simulate an arbitrary P machine [1].
- More fundamentally, a diagonalization proof would likely relativize, and Baker, Gill and Solovay showed no relativizable proof can settle the P versus NP problem in either direction [1]. The full citation is Baker, T., Gill, J., and Solovay, R., "Relativizations of the P=?NP question", *SIAM Journal on Computing* 4 (1975), 431 to 442 [3].
- A closely related oracle result is Bennett, C. H. and Gill, J., "Relative to a random oracle A, $P^A \neq NP^A$ with probability 1", *SIAM Journal on Computing* 10:1 (1981), 96 to 113 [3].

Circuit lower bounds and the natural proofs barrier

- It suffices to show some NP-complete problem cannot be solved by relatively small circuits of AND, OR and NOT gates, with gate count bounded by a fixed polynomial in the input size [1].
- Furst, Saxe and Sipser showed in 1984 that small circuits cannot compute parity at fixed depth; the full citation is Furst, M., Saxe, J. B., and Sipser, M., "Parity, circuits, and the polynomial time hierarchy", *Mathematical Systems Theory* 17 (1984), 13 to 27 [1][3].
- Razborov showed in 1985 that Clique has no small monotone circuits, that is, circuits with AND and OR gates but no NOT gates; extending that result to general circuits would prove $P \neq NP$, but Razborov later showed his own techniques fail miserably once NOT gates are allowed [1].
- Razborov and Rudich then developed the notion of "natural" proofs and gave evidence that the field's circuit-complexity techniques cannot be pushed much further; Fortnow adds, writing in 2009, that no significantly new circuit lower bounds had appeared in the preceding twenty years [1].

Arithmetization and the algebrization barrier

Aaronson and Wigderson's paper is the most technically specific source retrieved in full this run, and its abstract states the result directly [3].

- The starting observation: any proof of $P \neq NP$ must overcome relativization and natural proofs, yet the previous decade produced circuit lower bounds, for example that PP does not have linear-size circuits, that overcome both simultaneously, raising the question of a third barrier [3].
- Their barrier, algebrization or algebraic relativization, gives the simulating machine access not only to an oracle A but also to a low-degree extension of A over a finite field or ring [3].
- They show all known nonrelativizing results based on arithmetization do algebrize, including the inclusions $IP = PSPACE$ and $MIP = NEXP$ and the separation $MAEXP$ not-subset-of $P/poly$ [3].
- They show almost all the major open problems, including P versus NP, P versus RP, and NEXP versus P/poly, will require non-algebrizing techniques, and that in some cases algebrization explains exactly why progress stopped where it did, for example why superlinear circuit lower bounds exist for PromiseMA but not for NP [3].

What else the algebrization paper contains

The second half of the result is a lower-bound theory in its own right, which matters because it shows the barrier is proved rather than argued.

- The barrier results follow from lower bounds in a new model of algebraic query complexity introduced in the same paper [3].
- Some lower bounds use direct combinatorial and algebraic arguments; others come from a connection between that model and communication complexity [3].
- Using that connection the authors give an MA-protocol for Inner Product with $O(\sqrt{n} \log n)$ communication, essentially matching a lower bound of Klauck [3].
- They also state a communication complexity conjecture whose truth would imply $NL \neq NP$ [3].

Proof complexity

A fourth line of attack runs through the length of propositional proofs rather than circuits. If one could prove there are no short proofs of tautologies, that would imply $P \neq NP$; Armin Haken showed in 1985 that tautologies encoding the pigeonhole principle have no short resolution proofs [1]. But a separation needs the statement for an arbitrary proof system, and even a breakthrough showing tautologies have no short general Frege proofs would not suffice [1].

IMPLICATION: Each barrier is a theorem about what a proof can look like, so the honest summary of the state of play is that we know a great deal about how the problem cannot be solved and almost nothing about how it can.

3.4 What follows if $P = NP$, and the much weaker thing that follows if $P \neq NP$

The asymmetry here is the single most examinable point in the topic: $P = NP$ would be sweeping and constructive in its consequences, whereas $P \neq NP$ would leave essentially every practical security guarantee still unproved.

If $P = NP$

- Public-key cryptography becomes impossible [1].
- Fortnow argues the positive consequences dominate: since NP-complete optimization problems become easy, transport, manufacturing and scheduling become far more efficient, and learning becomes easy by applying Occam's razor, finding the smallest program consistent with the data, making vision recognition, language comprehension and translation trivial [1].
- Mathematics changes: proofs of theorems with reasonable-length proofs, say under 100 pages, become findable, so a person who proves $P = NP$ would collect not one Millennium prize but most of them [1].
- Fortnow attaches an explicit caveat that students routinely drop: "Technically we could have $P = NP$, but not have practical algorithms for most NP-complete problems" [1]. A proof could be nonconstructive, or the polynomial could be enormous.

If $P \neq NP$

The direction most people assume is the safe one delivers much less than they expect.

- Assuming $P \neq NP$ is not enough to get public-key protocols; instead one needs strong average-case assumptions about the difficulty of factoring or related problems [1].
- The implication chain runs one way only: public-key cryptosystems imply the existence of one-way functions, which in turn imply $P \neq NP$, and the retrieved MIT course notes state plainly that these are the only implications known, asking "Where does the truth lie?" [9].
- Even after a separation, cryptography would require that a problem like factoring, which is not believed to be NP-complete, is hard for randomly drawn composite numbers [1].
- Whether worst-case to average-case reductions of the kind known for lattice problems and the permanent also hold for NP-complete sets is called out as an important open problem [1].

Impagliazzo's five worlds

The gap between worst-case and average-case hardness is exactly what the five-worlds framework organizes. A retrieved 2026 arXiv preprint describes the framework as the most influential for reasoning about that gap and about the foundations of cryptography, and names the five worlds Algorithmica, Heuristica, Pessiland, Minicrypt and Cryptomania, defined by progressively stronger assumptions about the existence of computational hardness, running from a world where $P = NP$ to a world where public-key cryptography is possible [7].

That preprint's own contribution is a proposed sixth axis and is flagged separately below; its account of Impagliazzo's framework is background, and the primary source for the framework, Impagliazzo's 1995 paper, was not retrieved this run.

3.5 Where the research actually stands

The field's own retrospectives describe a problem that has not moved in decades, alongside a practice that has moved enormously without needing it to.

The stalled theory

- Fortnow's 2009 survey reports that circuit complexity and other approaches have stalled, with little reason to believe a separation will appear in the near future [1].
- His 50-year retrospective states that the problem turned 50 in 2021 and its resolution remains far out of reach, and that the theory itself has not changed dramatically since 2009 even though computing has [2].
- The longest-range programme named in the retrieved material is Geometric Complexity Theory, Mulmuley and Sohoni's attack through algebraic geometry, under which for each n an integral point in a certain high-dimensional polygon P_n would force any circuit family for Hamiltonian path to have size at least n to the log n on inputs of size n , implying $P \neq NP$; Mulmuley believes it will take about 100 years to carry the programme out, if it works at all [1].
- Quantum computing is not a way around it: factoring and discrete logarithm are not believed to be NP-complete, and Grover's algorithm, which does apply to general NP problems, achieves only a quadratic speed-up [1].

The unstalled practice

Fortnow's 2022 framing is that the interesting movement has been on the algorithmic side while the separation stayed put.

- Dramatic advances in algorithms and hardware have allowed the field to tackle many NP-complete problems while making little progress breaking cryptographic systems [2].
- Fortnow names the resulting regime "Optiland", a world where we almost miraculously gain many of the advantages of $P = NP$ while avoiding some of the disadvantages, such as breaking cryptography [2].
- He reframes the problem's role accordingly: no longer only a question about which problems are hard, but a lens for charting what is and is not possible for the future of learning and of the field [2].
- Concrete tractability figures reported in the 2009 survey include travelling salesperson instances of more than 10,000 cities solved in practice, and SAT solvers settling competition formulas of one million variables, against roughly 100 variables for the best general algorithms on hard 3SAT instances [1].

The independence question

A separate line asks whether the question is decidable from the usual axioms at all. Scott Aaronson wrote a Computational Complexity Column, edited by Lance Fortnow, titled "Is P Versus NP Formally Independent?", which considers whether $P = NP$ is independent of the usual axiom systems [10]. This run retrieved only a title-and-snippet preview of that column, so its actual conclusion is not established here; the file is served as `column81.pdf`, and the Bulletin of the EATCS issue number and year are [INFERRED - VERIFY] from that filename rather than read from the document.

3.6 Claimed proofs, and how to read one

A steady stream of claimed resolutions exists; the barriers are the standard against which they are judged, and a claim that does not address all three is not taken seriously.

This run retrieved one such claim in full: an SSRN preprint by Ararat Petrosyan, an independent researcher, titled "A Rigorous Proof of $P \neq NP$: Polynomial Construction of a Self-Referential Formula and Overcoming Complexity Barriers (Part II)", 7 pages, written 27 April 2025 and posted 7 May 2025 [5].

- The abstract claims a self-referential CNF formula refuting an "Incompressibility Hypothesis" for SAT, with a polynomial-time construction on a deterministic Turing machine replacing an earlier appeal to Kleene's fixed-point theorem [5].
- Its own framing is the useful part for a student: it argues explicitly that the method bypasses the Baker-Gill-Solovay relativization barrier and addresses the Razborov-Rudich and Aaronson-Wigderson barriers [5].
- SSRN is a preprint server, and the posting shows a sole self-funded author with 0 references fetched and 0 citations recorded [5]. Nothing in the retrieved record indicates peer review or acceptance at any venue.
- This brief takes no position on the claim's correctness and does not rely on it anywhere. It is cited only as documentation that claimed proofs circulate and that the three barriers are the checklist such claims are expected to clear.

REQUIRED ACTION: When you meet a claimed P versus NP proof, check first which of relativization, natural proofs and algebrization it addresses and how, before reading the technical body. A proof that does not explain why it escapes all three cannot be correct.

SECTION 04

4. COMMON MISCONCEPTIONS

Every entry below is documented by, or follows directly from, a statement in a retrieved source. Nothing here is a plausible-sounding student error invented for the table.

Misconception	What is actually true	Why the confusion arises
NP means "non-polynomial", so NP problems are by definition not in P.	NP abbreviates "Nondeterministic Polynomial-Time" [1], and P is contained in NP; whether the containment is strict is the open question [4].	The abbreviation reads like "non-P" in English, and the expansion is rarely stated in the same breath as the class is introduced.
$P \neq NP$ would make public-key cryptography secure.	Assuming $P \neq NP$ is not enough to get public-key protocols; strong average-case assumptions about factoring or related problems are needed [1]. The known implications run only the other way, from public-key cryptosystems to one-way functions to $P \neq NP$ [9].	NP-completeness is taught as worst-case hardness, and worst-case hardness is silently assumed to transfer to the random instances cryptography actually uses.
If $P = NP$, everything NP-complete becomes practically fast.	"Technically we could have $P = NP$, but not have practical algorithms for most NP-complete problems" [1].	"Polynomial" is taught as a synonym for "efficient", so a polynomial with a huge exponent or constant, or a nonconstructive proof, is never pictured.
NP-complete means hopeless in	Travelling salesperson instances of	NP-completeness is a worst-case

Misconception	What is actually true	Why the confusion arises
practice.	more than 10,000 cities are solved in practice, and SAT solvers settle competition formulas of a million variables [1]; decades of algorithmic advance have let the field tackle many NP-complete problems [2].	statement about one problem family, and courses rarely separate worst-case from typical-instance behaviour.
Factoring is NP-complete, which is why RSA is safe.	Factoring and discrete logarithm are not believed to be NP-complete [1].	Factoring is the canonical "hard problem" in a cryptography course and the canonical NP-complete problems come from an algorithms course, so the two lists get merged.
Quantum computers will solve NP-complete problems.	Shor's algorithms attack factoring and discrete logarithm, neither believed NP-complete, and Grover's algorithm, which does apply to general NP problems, gives only a quadratic speed-up [1].	Shor's algorithm is widely reported as "quantum computers break hard problems", and the class distinction is dropped in the retelling.
The barriers are evidence that $P \neq NP$.	The barriers are theorems about proof techniques, not about the answer; Baker, Gill and Solovay rules out relativizing proofs in either direction [1], and Aaronson and Wigderson show major open problems need non-algebrizing techniques without implying which way they resolve [3].	"We cannot prove X" and "X is false" are easy to conflate when the belief that $P \neq NP$ is already stated in the same paragraph.
Proving a circuit lower bound for one restricted model is nearly the whole job.	Razborov's monotone lower bound for Clique would imply $P \neq NP$ if extended to general circuits, but he later showed the technique fails miserably once NOT gates are allowed [1].	Restricted-model results are presented as steps on a ladder, which implies the remaining rungs are of the same kind.

SECTION 05

5. OPEN QUESTIONS AND WHAT TO ASK

- **1. Is there a non-algebrizing technique, and what would one even look like?** Aaronson and Wigderson show almost all the major open problems require such techniques [3] but the retrieved material does not identify a candidate. *Ask your instructor:* are there results since 2009, for instance in the "ironic complexity" line connecting algorithms to lower bounds, that are known to evade all three barriers?
- **2. Do worst-case to average-case reductions hold for NP-complete sets?** Fortnow names this an important open problem, noting such reductions are known for lattice shortest-vector and the permanent, neither believed NP-complete [1]. *Ask your instructor:* which of Impagliazzo's five worlds does the current evidence most constrain, and why is Pessiland so hard to rule out?
- **3. Which of the five worlds do we live in?** The framework runs from Algorithmica, where $P = NP$, to Cryptomania, where public-key cryptography is possible [7], and the only implications known are

that public-key cryptosystems imply one-way functions which imply $P \neq NP$ [9]. *Ask your instructor:* is Impagliazzo's 1995 paper assigned reading, and which world does the course treat as the working assumption?

- 4. **Is P versus NP formally independent of standard axioms?** Aaronson wrote a full column on the question [10], but this run recovered only a preview, so the argument and its conclusion are not established here. *Ask your instructor:* is independence considered a live possibility or a curiosity, and what would be required to establish it?
- 5. **Can Geometric Complexity Theory's remaining steps be carried out?** The programme reduces a lower-bound question to the existence of integral points in a family of algebro-geometric polygons, and its own originator estimates roughly 100 years [1]. *Ask your instructor:* has GCT produced any unconditional complexity-theoretic consequence yet, or is its value still entirely prospective?
- 6. **What exactly does the official Clay problem description say, and how does it differ from the textbook statement?** The Clay description by Stephen Cook is the source of record and was not opened in this run (Section 7). *Ask your instructor:* should the course's definitions be lined up against Cook's official formulation, particularly the nondeterministic-machine phrasing versus the verifier phrasing?

SECTION 06

6. METHOD, SCOPE AND SOURCE BASIS

6.1 What was researched and how

This brief rests on eleven sources retrieved and selected mechanically, of which seven were read as scraped text and four were available only as title-and-snippet previews.

- Five approved sub-queries were searched, covering the formal statement, Cook-Levin, the proof barriers, the consequences for cryptography, and the current research status.
- Candidate URLs were ordered by a six-level authority ladder rather than by apparent relevance, and a floor-and-ceiling rule then selected 11 sources from 50 candidates against a ceiling of 25.
- Selected HTML pages were fetched and their main content extracted; PDFs were never fetched, by design, and are carried as title-and-snippet previews only.
- Each source is persisted as an excerpt capped at 3,000 characters. The live scrape of source 1 returned the complete article, so a minority of the detail cited to [1], specifically the proof-complexity, practical-tractability, quantum and Geometric Complexity Theory passages, comes from that full retrieval and sits beyond the persisted 3,000-character excerpt of the same page.

6.2 Reliability basis and precedence

Where sources differ in authority this brief follows the higher tier and says so; two of the three technically load-bearing sources are peer-reviewed, and no claim rests on a single low-tier source alone.

- Source 3, Aaronson and Wigderson in *ACM Transactions on Computation Theory*, is peer-reviewed and is the authority for everything said about algebrization and for the full bibliographic details of Baker-Gill-Solovay, Bennett-Gill and Furst-Saxe-Sipser.

- Sources 1 and 2, Fortnow in *Communications of the ACM*, are signed survey articles by a complexity theorist in the ACM's flagship publication; they are the authority for the historical narrative, the barriers as a story, and the practical-tractability figures.
- Source 4, IEEE Technology Navigator, is an editorially curated reference page rather than a peer-reviewed article. It is used only for definitional material that sources 1 and 3 independently corroborate, never as a sole authority for a technical claim.
- Sources 5 and 7 are unrefereed preprints and are labelled as such at every point of use. Source 5's claimed proof is not relied on anywhere; source 7 is used only for its restatement of Impagliazzo's five-world framework, not for its own proposed extension.

6.3 Notation and comparison conventions

The brief matches the notation of the sources it cites and flags the one place where the sources use two different names for the same object.

- Class names are written in plain capitals (P, NP, PSPACE, NEXP, P/poly) because the retrieved sources render them variously in bold, sans-serif and TeX markup with no substantive difference.
- Negation is written `!=` throughout, matching the plain-text rendering in which the sources were retrieved. Your course will almost certainly write it with a struck equals sign.
- Aaronson and Wigderson use "algebrization" and "algebraic relativization" interchangeably for the same barrier [3]; this brief uses "algebrization" and notes the synonym in Section 2.
- Dates are given as the sources state them. Where two retrieved sources date the same event differently in precision, both are given, as with Cook's 1971 presentation.

6.4 Gaps, caveats and what this brief does not establish

The most significant limitation of this run is that the single most authoritative document for the question, Cook's official Clay Mathematics Institute problem description, was identified but never opened.

- **CONTEXT CONFLICT:** The supplied context states that "Primary sources, published papers, and university course notes are preferred over general web explainers; PDFs are explicitly in scope." This pipeline never scrapes PDFs; every PDF is carried as a title-and-snippet preview. Four of eleven sources, including the MIT course notes and Aaronson's independence column, are therefore preview-only, and the stated preference for PDFs could not be honoured as retrieved evidence.
- **CONTEXT CONFLICT:** The same stated preference for primary sources was partly defeated by the authority ladder itself, which does not classify ``claymath.org`` or ``scottaaronson.com`` and defaulted both to the general-web tier. Cook's official Clay problem description and Aaronson's 116-page survey were both excluded from the fetch set for that reason, not on merit. Both are listed in Section 7 and both domains are reported for ladder extension.
- No claim in this brief is drawn from general knowledge without a tag. The only untagged claims are those carrying an inline citation to a retrieved source; the one inferred item, the Bulletin of the EATCS issue number for source 10, is tagged **[INFERRED - VERIFY]** at the point of use in Section 3.5.
- This brief has not been independently fact-checked. Every URL was verified against what this run actually retrieved, but nothing verifies that a correctly cited URL has been attached to the right claim, and that is a different failure class which this pipeline does not cover.

SECTION 07

7. SOURCES

Sources Cited

Eleven sources, numbered to match the inline citation links; entries 8 through 11 were available as title-and-snippet previews only and are also listed in the following subsection.

- 1. **The Status of the P Versus NP Problem.** Lance Fortnow. *Communications of the ACM*, Vol. 52 No. 9 (September 2009), pages 78 to 86. DOI 10.1145/1562164.1562186.
<https://cacm.acm.org/research/the-status-of-the-p-versus-np-problem/>
- 2. **Fifty Years of P vs. NP and the Possibility of the Impossible.** Lance Fortnow. *Communications of the ACM* (published following the problem's 50th anniversary in 2021).
<https://cacm.acm.org/research/fifty-years-of-p-vs-np-and-the-possibility-of-the-impossible/>
- 3. **Algebrization: A New Barrier in Complexity Theory.** Scott Aaronson (MIT) and Avi Wigderson (Institute for Advanced Study). *ACM Transactions on Computation Theory*, Volume 1, Issue 1 (February 2009). DOI 10.1145/1490270.1490272. Refereed research article.
<https://dl.acm.org/doi/10.1145/1490270.1490272>
- 4. **NP-complete problem.** IEEE Technology Navigator, curated by IEEE. Editorially curated reference page, undated. <https://technav.ieee.org/topic/np-complete-problem/>
- 5. **A Rigorous Proof of $P \neq NP$: Polynomial Construction of a Self-Referential Formula and Overcoming Complexity Barriers (Part II).** Ararat Petrosyan, Independent Researcher. SSRN preprint 5232844, 7 pages, written 27 April 2025, posted 7 May 2025. DOI 10.2139/ssrn.5232844. Unrefereed preprint; claim not endorsed here. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5232844
- 6. **The P versus NP Problem.** Rakesh Dube. arXiv:1001.3816 [cs.CC], submitted 21 January 2010, withdrawn 22 January 2010 by arXiv administration for plagiarism of Stephen Cook's Clay Mathematics Institute problem description. Cited only for that administrative record.
<https://arxiv.org/abs/1001.3816>
- 7. **The Observer World: A Cryptographic Extension of Impagliazzo's Five Worlds.** Fabio Francesco Gabriele Buono, Independent Researcher. arXiv:2606.27139v1 [cs.CR], 25 June 2026, CC BY 4.0. Unrefereed preprint; used only for its restatement of Impagliazzo's framework.
<https://arxiv.org/html/2606.27139v1>
- 8. **Algebrization: A New Barrier in Complexity Theory (author copy).** Scott Aaronson and Avi Wigderson, hosted at the Institute for Advanced Study. Preview only, not full text.
<http://www.math.ias.edu/~avi/PUBLICATIONS/MYPAPERS/SCOTT/alg.pdf>
- 9. **Lecture 23: Average-Case Complexity.** MIT course scribe notes hosted under Madhu Sudan's CSAIL pages; the course identifier ST07 is taken from the URL path. Preview only, not full text.
<https://people.csail.mit.edu/madhu/ST07/scribe/lect23.pdf>
- 10. **Is P Versus NP Formally Independent?** Scott Aaronson. The Computational Complexity Column, edited by Lance Fortnow, Bulletin of the EATCS; served as `column81.pdf` from a University of Ulm mirror. Issue number and year [INFERRED - VERIFY]. Preview only, not full text. https://www.uni-ulm.de/fileadmin/website_uni_ulm/iui.inst.190/Mitarbeiter/toran/beatcs/column81.pdf
- 11. **On the Boolean Satisfiability.** HAL preprint hal-04868914v3. Preview only, not full text.
https://hal.science/hal-04868914v3/file/Boolean_Satisfiability_.pdf

Identified but Not Retrieved in Full

Six documents were found and judged useful but never read end to end, and the two that matter most are the primary source of record and the field's standard modern survey.

Locations below are given without a clickable link because this run never retrieved these documents, and this pipeline's citation check admits only URLs it actually fetched. They are ordered by tier, most authoritative first.

- **1. The P versus NP problem, Stephen Cook, official Clay Mathematics Institute problem description** (PDF at ``claymath.org/wp-content/uploads/2022/06/pvsnp.pdf``). **This is the primary source of record for the formal statement of the problem** and the single most important document for this query. It was excluded from the fetch set because the authority ladder does not classify ``claymath.org`` and defaulted it to the general-web tier, not because it was judged unimportant. Everything Section 3.1 says about the formal statement therefore rests on secondary accounts.
- **2. P =? NP, Scott Aaronson, survey article** (PDF at ``scottaaronson.com/papers/pnp.pdf``). **The standard modern survey of exactly this question.** Retrieval previews describe it as a 116-page survey published in **Open Problems in Mathematics** (Springer), covering diagonalization, circuit lower bounds, the relativization, algebrization and natural proofs barriers, and the Williams and Mulmuley programmes; page count, venue and scope are **[INFERRED - VERIFY]** from those previews and were not confirmed against the document itself. Excluded for the same ladder reason, ``scottaaronson.com`` being unclassified. Its absence is the largest single gap in this brief's coverage of the post-2009 research status.
- **3. Algebrization author copy** (tier 2, university and research-institute hosting, at ``math.ias.edu``). Preview only, because this pipeline does not scrape PDFs. The peer-reviewed version of the same paper was retrieved in full as source 3, so this gap is immaterial.
- **4. Lecture 23: Average-Case Complexity, MIT course scribe notes** (tier 2, university course material, at ``people.csail.mit.edu``). Preview only, because this pipeline does not scrape PDFs. The one sentence recovered from the preview is load-bearing in Section 3.4 and should be checked against the full notes.
- **5. Is P Versus NP Formally Independent?, Scott Aaronson, Computational Complexity Column** (tier 2, university mirror at ``uni-ulm.de``). Preview only, because this pipeline does not scrape PDFs. Its conclusion is not established anywhere in this brief.
- **6. On the Boolean Satisfiability, HAL preprint hal-04868914v3** (tier 3, preprint server, at ``hal.science``). Preview only, because this pipeline does not scrape PDFs. Cited once, for a single sentence about 3-SAT, and nothing important depends on it.